```
!pip install nltk
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.1)
```

```
import nltk

nltk.download("brown")
nltk.download("universal_tagset")
```

```
[nltk_data] Downloading package brown to /root/nltk_data...
[nltk_data]   Unzipping corpora/brown.zip.
[nltk_data] Downloading package universal_tagset to /root/nltk_data...
[nltk_data]   Unzipping taggers/universal_tagset.zip.
True
```

```
from nltk.corpus import brown

sentences = brown.sents()
tokens = [word.lower() for sent in sentences for word in sent]

print("Total sentences:", len(sentences))
print("Total tokens:", len(tokens))
print(tokens[:20])
```

```
Total sentences: 57340
Total tokens: 1161192
['the', 'fulton', 'county', 'grand', 'jury', 'said', 'friday', 'an', 'investigation', 'of', "atlanta's", 'recent', 'primary', 'e
```

```
from nltk.tag import hmm

train_data = brown.tagged_sents(tagset="universal")

trainer = hmm.HiddenMarkovModelTrainer()
hmm_model = trainer.train_supervised(train_data)

print("HMM Model Trained Successfully")
```

```
HMM Model Trained Successfully
```

```
from collections import defaultdict, Counter

pos_sequences = [[tag for (_, tag) in sent] for sent in train_data]
pos_bigram = defaultdict(Counter)

for seq in pos_sequences:
    for i in range(len(seq) - 1):
        pos_bigram[seq[i]][seq[i + 1]] += 1

word_by_tag = defaultdict(Counter)

for sent in train_data:
    for word, tag in sent:
        word_by_tag[tag][word.lower()] += 1
```

```
def hmm_predict_next(sentence):
    words = sentence.lower().split()

    tagged = hmm_model.tag(words)
    last_tag = tagged[-1][1]

    next_pos = pos_bigram[last_tag].most_common(1)[0][0]

    candidates = word_by_tag[next_pos].most_common(5)

    return next_pos, candidates
```

```
pos, words = hmm_predict_next("the government decided")
```

```
print("Predicted POS:", pos)
print("Suggested Next Words:", words)
```

```
/usr/local/lib/python3.12/dist-packages/nltk/tag/hmm.py:335: RuntimeWarning: overflow encountered in cast
  O[i, k] = self._output_logprob(si, self._symbols[k])
Predicted POS: VERB
Suggested Next Words: [('is', 10108), ('was', 9815), ('be', 6374), ('had', 5132), ('are', 4394)]
```